

5(a)	any two from: cells secrete ligands or transport of ligands, through blood / in circulatory system ; ligands bind to (complementary / specific) <u>receptors</u> ; <i>context of target cells</i> (binding) sets off reactions within the (target) cell / triggers secondary messengers / sets off an enzyme cascade ; A cyclic AMP for secondary messengers	2
5(b)	(non-competitive inhibitor) binds to, allosteric site / site other than active site / AW ; (binding causes) change in shape of active site ; active site no longer complementary to the substrate / AW ;	2
5(c)(i)	Palbociclib / p21Cip1 ; (in S phase), stops DNA replication occurring / DNA replication occurs or identical /sister, chromatid formed in S phase ; A <i>idea of</i> p21Cip1 disrupts normal DNA replication (in S phase)	2

Question	Answer	Marks
5(c)(ii)	<u>RO-3306</u>, <i>qualified</i> 1 cell cycle stops (in G₂) after mitochondria have, divided / increased in number or mitochondria, divide / increase in number, in (G1 and) G2 / before mitosis ; A no, mitosis / cytokinesis, so numbers of mitochondria remain high 2 <i>idea that</i> DNA replication will have occurred / sister chromatids will have formed, in S phase, but the cell cannot carry out mitosis ;	2
5(d)	cell cycle is stopped before, mitosis / cytokinesis / cell division ; <i>idea of</i> stops uncontrolled cell division (increasing tumour size) ;	2